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Wildland Fire Prevention: a Review

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Abstract

Purpose of Review Prevention effectiveness relies on knowledge about the spatio-temporal likelihood of fire occurrence and whether fires are a result of natural processes or human influence. Such knowledge can be used to develop more effective tools and strategies to reduce both the likelihood of and damage from unwanted fires. The purpose of this paper is to provide a review of wildland fire prevention, which pertains to any actions used to reduce the likelihood and damage from unwanted wildfires.

Recent Findings Research suggests that there have been significant advances in developing models and approaches to estimating the likelihood of wildfire occurrences. Similarly, research in the social sciences is advancing knowledge regarding human behavior and factors that will lead to enhanced wildfire prevention efforts.

Summary However, questions asked in the 1960s regarding the links between attitudes and beliefs and how changes in such result in decreased fire occurrence remain tenuous. Continuing interdisciplinary work is required to promote a deeper understanding of what factors will influence human behavior to make wildfire prevention programs more effective. Although the fire prevention literature pertains predominantly to North America, I have tried to include both studies and policies from other fire-prone jurisdictions.

Keywords Wildfire likelihood · Human-caused fires · Wildfire prevention education · Risk perception

Introduction

The phrase “Only YOU can prevent forest fires!” elicits images of an iconic bear in North America, and indeed, worldwide. Since 1944, Smokey Bear has been urging the public in the longest-running public service campaign to take care in the forest in order to prevent wildfires.¹ Recognizing the need to change human behavior, William Folkman, a sociologist and the project leader for fire prevention research at the USDA Forest Service’s Pacific Southwest Forest and Range Experiment Station based in California, conducted many studies in the 1960s and 1970s that focused on the public’s knowledge of fire risk and prevention. He was particularly interested in the socio-economic characteristics related to fire starts for a

wide range of the population including hunters [1], recreation users [2], children [3], motorists [4], and youth [5]. He concluded that to make a difference, it would be necessary to understand the problem rather than treat just the symptom and that prevention messaging and education must target the full spectrum of both urban and rural populations.

Over the same time, a successful wildfire suppression strategy undertaken by federal, state, and local governments resulted in an increase in vegetation available to burn in future wildfires, termed “hazardous fuels,” but more importantly, institutionalized a paradigm whereby the use of suppression and control became the predominant management strategy to the exclusion of other approaches. This is true in North America as well as the Mediterranean, for example. Notwithstanding, researchers at the time suggested that fire planning must still focus on prevention as an essential part of operations, and that fire management goals must be consistent [6, 7].

The 1994, fire season in North America was particularly devastating in terms of the loss of firefighters’ lives, thus eliciting a response from the National Wildfire Coordination Group (NWCG) in the USA to return to prevention as a more predominant protection strategy. The group released a report [8] focused on prevention that characterized the main causes of wildland fires and potential prevention strategies.

¹ Smokey Bear even has his own website: <https://smokeybear.com/en>

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Since then, there has also been a shift in wildland fire policy to include the use of wildfire to achieve management objectives, mitigate fire hazards on private property and in communities, and to emphasize community and privately driven prevention strategies. Research has followed suit with a return to greater emphasis on the social sciences in pursuit of understanding human behavior as it relates to wildfire prevention.

While the worldwide fire problem continues to grow in complexity as a result of human development and ecosystem change [9], fire management objectives continue to focus on protecting highly valued resources and assets. Bradshaw [10] described the problem as being one of contrasts whereby fire management now requires expertise regarding protection of both structures and wildlands, where ownership crosses legal boundaries, and where responsibility for protection is not always clear. How do wildland fire managers balance the need for fire as a natural and necessary ecological process, while at the same time prevent unwanted wildfires that threaten both anthropocentric values and ecosystem functioning?

Prevention begins with an understanding of when and where fires are more likely to occur, and why they ignite. Following this logic, fire prevention strategies can be designed to directly target the source of the problem. A prescient image presented in Fire Management Notes in 1972 [11] spoke to “new dimensions” of fire prevention, namely fuel management (Fig. 1). It is interesting to note that while fuels treatment was being touted as a new opportunity, that today education, law enforcement, and modification of human behaviour remains equally important.

Education, community collaboration, and media campaigns have been used worldwide to increase knowledge and disseminate information to the appropriate audiences in time and space. The question is, to what extent have these

strategies been effective in reducing the likelihood of fire occurrence and potential damage? What characteristics in the delivery and timing of such programs lead to greater efficiency?

This paper reviews the wildland fire prevention literature. While there are many fire prone jurisdictions worldwide, the vast majority of fire prevention research and literature pertains to North America. I have included studies from Europe (largely the Mediterranean) and Australia where pertinent and available. The review begins with spatio-temporal approaches to predicting the likelihood of wildfire occurrences, followed by a review of educational strategies and regulatory approaches used to prevent unwanted wildfires. Lastly, I look at literature that attempts to evaluate the efficacy of fire prevention strategies, including research that investigates the characteristics of how such strategies are delivered. I conclude with a discussion describing what has been learned, and where gaps remain.

Likelihood of Wildland Fire – Where, when, and why

The NWCG [8] classified fire causes into nine categories only one of which is due to natural sources (e.g. lightning, volcanic). The remaining categories are a result of human influence and include: campfires, smoking, fire use, incendiary, equipment, railroads, juveniles, and miscellaneous. Since then, Prestemon et al. [12] developed a conceptual model of prevention that differentiates between three broader categories of fire causes: natural, and anthropocentric, the latter of which includes accidental and incendiary. Several researchers have examined fire ignition patterns including location and timing to determine how better to target fire prevention strategies.

Early fire prevention research by Christiansen et al. [13] recognized the importance of human behaviour in addition to biophysical factors related to wildfire ignitions. They provided a series of recommendations based on location (hotspots), demographics, and social considerations such as attitudes and beliefs, which they suggest influence human behaviour. They concluded that such information could be used to develop long-term strategies and to monitor and evaluate progress.

Along the same lines, Thomas et al. [14] recommended the need for three types of information to improve the efficiency of wildland fire prevention efforts: (i) the characteristics of wildfire data including the cause and extent of fire occurrence, (ii) spatio-temporal ignition patterns, and (iii) the effects of prevention activities. Having information about each can improve the use of limited resources and more efficiently achieve long-term prevention goals. Their results showed that fires (both intentional and accidental) occur in spatial and temporal clusters making them somewhat predictable. Understanding where and when fires are more likely to occur within the landscape can help tailor activities that are delivered in

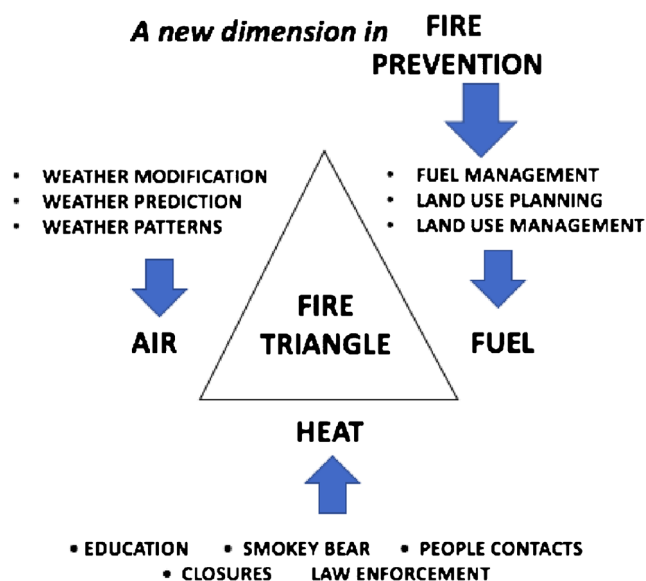


Fig. 1 Prevention activities remain as important as ever in spite of advances in fuels management. (Source Fire Management Notes, 1973)

geographically relevant locations, to targeted audiences, and at appropriate times [15]. Additionally, such knowledge provides insight into where and when to target information campaigns, and when to increase law enforcement.

While natural causes are responsible for many wildland fires, the majority worldwide are human-caused fires (HCFs). Furthermore, human influence has increased both the seasonal length and geographical extent of wildfires as per wildland fire statistics in the U.S. [16]. All such statistics suggest that although there are many individuals and communities that understand wildfire and the associated risks, more needs to be done to educate those who do not. Estimating the exact ratio of causes is difficult because percentages vary annually by country and location. Additionally, many ignition sources are unknown and cannot be categorized. Nevertheless, in some jurisdictions HCFs represent a significant ignition source. Nagy et al. [17••] redefine “large” fires as the biggest 10% by ecoregion rather than fire size threshold. They illustrate clearly that proportionally, humans started the majority of large fires in the U.S.: 92% in the East and 65% in the West. South [18] reported that 90% of all fires in 2016 in the U.S. were due to human activity. The University of Colorado at Boulder reported that 84% of wildfires occurring in the U.S. and Canada were HCFs.² The analysis used information from the Fire Program Analysis Fire-Occurrence Database (FPA-FOD). Data were categorized as natural cause (lightning), human-related causes (10 different categories), and miscellaneous and missing/undefined for instances where fire ignition source was unknown.

Tedim et al. [19] examined forest fire causes in Europe, also concluding that the majority of ignitions were due to human factors. They discuss how fires in the 1930s were mostly due to negligence related to industrial and commercial activity. Today, humans cause the majority of wildland fire ignitions either accidentally (most of Europe) or intentionally (Spain, Italy, and Portugal). Hirshberger [20] provides greater detail about arson fires in Europe indicating that of the 76% of fires of known cause in Italy, 85% are a result of arson. The case is similar for Spain: 59% out of 88% of fires of known cause were incendiary, and in Portugal, 31% out of 62% of fires of known cause were incendiary.

Early work on incendiary fires in the USA took a three-pronged approach. Riebold [21] borrowed from a highway safety program known as the three E’s—education, enforcement, and engineering—to view fire prevention possibilities, particularly related to arson. He recommended reclassifying the setting of incendiary fires as a felony offense with more severe penalties and a greater reliance on law enforcement. He also noted that education must be combined with enforcement to change the public’s behavior. To aid in prevention efforts, it

is further important to use engineering solutions such as forest thinning and prescribed burning to reduce hazards. Doolittle et al. [22] also focused on prevention of incendiary fires in the southern USA and suggested a step-wise strategy that addresses the process from an arsonist’s motivation to future prevention.

Donoghue and Paananen [23] also recognized the importance of the legal system as a means to help reduce HCFs. They contended that the system should be used in conjunction with education; however, as effectiveness requires that fire managers understand the legal system and know how to use it, their paper aimed to describe the legal system in the context of wildfire offenses.

Wildfire ignition in Canada does not reflect as much human influence given the sparse population distribution, particularly in the northern regions of the country. According to Environment Canada,³ wildfires started by lightning represent 45% of all fires, meaning that 55% are human-caused. In spite of this statistic, unwanted human ignitions are responsible for tremendous socio-economic damage and remain a focus of fire prevention programs, particularly in the wildland urban interface (WUI)—an area where human development intermingles with forested landscapes.

In Australia, the majority of bushfires are due to human influence with only 6% produced by lightning. Statistics show that arson is responsible for 13% of fires, and suspected arson, 32%. The remainder is largely a result of negligence. The same holds true in Russia with 72% a result of arson or carelessness. In the Amazon basin and in Indonesia, wildfires are almost exclusively due to human influence through land clearing and careless forestry practices. In both regions, lightning is generally accompanied by heavy rain and therefore natural fire causes are almost non-existent [20].

Modeling when and where wildfire is likely to occur from natural causes is a function of weather [24, 25]; ecosystem characteristics such as drought [26, 27] and insects/pathogens [28], and in the longer term, climate [29–31]. Rather than provide a review here of the many advances in methods and models to predict the likelihood of wildfire due to natural disturbances, the reader should refer to Seidl et al. [32] who provide a review of natural disturbance models for forest ecosystems that includes a comprehensive look at wildfire. They discuss a range of modeling approaches to evaluate ecosystem susceptibility and wildland fire occurrence that includes stochastic and deterministic, and spatio-temporal approaches. The authors review interactive effects of many disturbance agents including storm events, insects, pathogens, and climate change. Their primary recommendation is for future research to consider natural effects more holistically to more accurately model disturbance. They suggest that having

² Press release: <https://phys.org/news/2017-02-humans-percent-wildfires-season-decades.html> Accessed May 18, 2018.

³ Forest fires and lightning: <https://www.canada.ca/en/environment-climate-change/services/lightning/forest-fires.html>. Accessed May 18, 2018.

a broader understanding of how natural effects are interrelated could enhance fire prevention strategies with a specific aim to reduce the likelihood of wildland fire occurrence.

In summary, there is a tremendous body of literature devoted to spatio-temporal and socio-economic factors related to the occurrence of HCFs. Research estimating the likelihood of fire timing and geographic location by cause is extensive. Woodard and Niederleitner [33] looked at data from the 10 Canadian provinces that identified for each wildfire, the responsible party, and general and specific fire causes. They concluded that data failed to provide sufficient information to develop effective fire prevention programs. They recommended that data collection should focus on information regarding human behavior, specifically, unsafe acts.

In a recent paper, Costafreda-Aumedes et al. [34••] provide an expansive review of modeling developments aimed specifically at predicting HCFs. Their scan of nearly 200 papers considers 181 models and spans all global regions to identify the most appropriate variables and effective methods to model the likelihood of wildfire including those resulting from arson. They organize their work according to spatio-temporal dimensions and model outputs. Their findings indicate that while there are many promising models to estimate both short- and long-term effects, such models have largely not been adopted to inform fire management. Additionally, data from the majority of these studies are not available, thereby preventing comparative analyses to further strengthen predictive capacity.

Fire Prevention Controls—Education and Regulation

Responsibility for wildland fire prevention formally resides with public agencies, and less formally through not-for-profits, community organizations, and emerging partnerships involving key stakeholders for example. Education campaigns and public service announcements are used in many jurisdictions and are aimed at communicating knowledge and practices that can reduce the likelihood of wildfire ignitions. They include messages broadcast using a variety of traditional media (radio, television), social media (Facebook, Twitter), agency websites, print materials, and school campaigns for example. Messages focus on everything from proper campfire use (“soak it, stir it, soak it again”), to information regarding recreation (e.g., sparks from ATVs and motorcycles), appropriate use of power tools (e.g., refueling hot equipment), and debris burning on private land.

Information can be disseminated to the public to build knowledge around wildland fire policies and fuels management activities on public lands, as well as to increase the awareness of risk from wildland fire and strategy or program effectiveness. Much research has gone into developing and identifying the role of educating the public to promote

prevention knowledge. Many educational programs are designed not just to provide information about fire prevention strategies, but also to influence participants’ perceptions about wildland fire risk, attitudes regarding different prevention strategies, and beliefs about effectiveness.

Smokey Bear (Fig. 2) has been the longest-running ad campaign in history having served since 1944 at which time he was created to raise awareness about wildfire prevention [35]. The campaign has been successful in terms of message exposure, increasing awareness, and engagement. Robertson in a special issue of Forest Management Notes dedicated to Smokey [36] attributed stakeholder cooperation as one of the key components to the campaign’s success. Donated media contributions to the campaign in 2013 amounted to over \$34 M, and about 70% of adults recalled Smokey Bear’s message without prompting. The Ad Council notes that while there has been a substantial decline in the number of wildfires in the last 70 years, it is not possible to determine to what extent this decline is a direct result of the campaign.

Since Smokey Bear’s inception, fire management has changed from eliminating fire from the landscape to recognizing its ecological importance within many forest regions. Donovan and Brown [37] speak to Smokey’s legacy and suggest that prevention messaging needs to be broadened. Specifically, that more fire on the landscape is necessary to result in greater long-term gains.

As the WUI has become more extensive, and because HCFs account for the majority of fire starts, fire prevention has grown to include more strategies that focus on treating private property. Examples of engineering-type approaches



Fig. 2 Smokey Bear roadside wildfire prevention sign. (Source CC0-<https://pixabay.com/en/smokey-bear-fire-prevention-857181/>)

are Firewise and FireSmart: programs used in the USA and Canada, respectively, to reduce the likelihood of fire on private property, and in the event of fire, to prevent it from spreading [38]. Both programs are founded on physical science demonstrating that vegetative and structural modification can significantly reduce the likelihood of ignition or fire spread. Firewise⁴ was developed by the National Fire Protection Association (NFPA) in the USA in the 1980s in response to unprecedented structural losses in the WUI as a result of wildfire. The program began by increasing awareness of the risks associated with wildfire, and then provided specific guidance on how to mitigate risk on private property. Recommendations focus on landscaping with fire-resistant vegetation and removing flammable materials. The formal “Firewise” program was launched in 1997. The program provides information to WUI residents about safety in the home ignition zone [39] based on research from the International Crown Fire Modelling Experiment in the Northwest Territories, Canada [40].

The Canadian program, FireSmart, is modeled on the same set of principles and is used to reduce the susceptibility of communities and private property to fire risk. FireSmart was first initiated in Alberta in the late 1990s by Partners in Protection (PiP) to create awareness and workable solutions for vulnerable interface communities. Recommended prevention activities include landscaping and modification/reduction of fuel loads within the home ignition zone [41, 42].

Community Wildfire Protection Plans (CWPPs) are another approach to wildfire prevention for communities. CWPPs arose formally out of the Healthy Forests Restoration Act (HFRA) in the USA as a way for communities to influence the location and timing of public fuels reduction projects [43]. The strategy is to develop priorities for wildfire preparedness and fuels reductions. Less formally, community protection plans are used throughout Canada to educate communities and to identify needs based on risk and available resources. There are many templates available that describe essential requirements to developing CWPPs including descriptions of the local area and community, values at risk, wildfire threat, and options for action. For examples of templates and processes, see the Union of BC Municipalities⁵ and the International Association of Fire Chiefs.⁶

Australia uses a range of programs including Community Fireguard (CFG) in Victoria, Community FireSafe in South Australia, Community FireWise in New South Wales, and Bushfire Ready Action Groups (BRAG) in Western Australia. The group-based community approach focuses on

educating and training individuals on a prescribed set of protection and preparedness principles or on more custom needs, depending on the program. In each program, trained facilitators educate participants with the expectation that participants will adopt necessary actions in the future [44].

Turco et al. [45] review wildfire trends in Europe and state that wildland fire management and research is mostly about suppression and fuels management, with a call to action for more work to be done in the WUI. The International Association for Mediterranean Forests [46] released a position paper following two working group meetings regarding wildland fire prevention. The paper provides five recommendations to heighten wildfire prevention considerations: (1) enhance international cooperation, (2) integrate wildfire prevention into forestry programs and policies as well as climate change strategies, (3) promote knowledge and education, (4) enhance sustainable financial mechanisms, and (5) harmonize information systems to deal with wildfire risk. Of note is that recommendations do not include research.

Scant literature also suggests that there are few prevention programs aimed at individuals and communities. This might be the case given different legal systems, land-ownership, and legislated responsibilities for wildfire mitigation in many European countries [47]. Similarly, there is also a dearth of literature pertaining to wildfire prevention in other fire-prone jurisdictions in South and Central America, Africa, and Asia.

Technical approaches to wildfire prevention have generally relied on engineering solutions with a strong emphasis on regulatory compliance with respect to equipment use and permitted operations. One of the most prominent causes of wildfire in North America in the early to mid 1900s was from railroads. Pottharst et al. [48] used case studies of railroad-caused fires to look at the relationships between engineering solutions and ignition reduction. They found that improved products could reduce wildfire occurrence and that maintenance could support future reductions.

Grunstra and Martell [49] examined railway fires in Ontario and demonstrated how fires have been significantly reduced as a result of advances in technology (the move from coal-fired locomotives to diesel, spark arrestors, and cast-metal brake shoes), better fuels management in close proximity to rail lines (fuel reduction in the right-of-way), monitoring along rail lines and enhanced overall fire detection, and policy (e.g., regulations regarding equipment standards and fire safety devices).

Policy related to railway operations and wildfire prevention is easily accessible on government websites. For example, Canada’s Railway Safety Act (P.C. 2016-1154)⁷ defines fire hazard and risk followed by regulations pertaining to fire control, preparedness and planning, hazard reduction,

⁴ See Firewise at NFPA: a brief history. <http://www.firewise.org/about/history.aspx> (accessed December 19, 2017)

⁵ UBCM, retrieved from: <https://www.ubcm.ca/EN/main/funding/lgps/strategic-wildfire-prevention/2017-swp-program.html>

⁶ International Association of Fire Chiefs, retrieved from <https://www.iafc.org/topics-and-tools/resources/resource/community-guide-to-preparing-and-implementing-a-community-wildfire-protection-plan>

⁷ Prevention and Control of Fires on Line Works Regulations, retrieved from <http://www.gazette.gc.ca/rp-pr/p2/2016/2016-12-28/html/sor-dors317-eng.html>. Accessed May 18, 2018

communications, and maintenance and equipment inspection. The Government of California produced a field guide⁸ that defines standards and practices based on research and experience that is expected to minimize the likelihood of wildfire occurrences. The guide provides information regarding equipment use, safety practices, rights of way, and statutes and regulations for example.

Rail companies in Australia are also taking the initiative to prevent bushfires. The Australian Rail Track Corporation (ARTC)⁹ produces a fact sheet for communities explaining their preventative measures including equipment maintenance, vegetation clearance, safety inspections, and education of staff. They also highlight technical solutions to help reduce the likelihood of fire including trackside monitoring across the rail network.

Regulations to prevent wildfires resulting from industrial activity are critical in fire-prone jurisdictions and involve both technological and regulatory approaches. Powerlines are considered to be one of the leading causes of wildfires in several regions of the globe and use and maintenance is highly regulated. This includes equipment safety standards, inspections and maintenance, and vegetation clearance. An example of a technological approach to powerline-caused wildfires is the Texas Wildfire Mitigation Project, funded by the Texas legislature. The Texas A&M Engineering Experiment Station developed monitoring technology to detect downed power lines, line equipment failure, and arcing equipment, all of which can cause wildfires. The hope is to prevent wildfires and increase response time through enhanced detection [50].

The same is true for industries such as oil and gas, mining, and forestry, all of which undertake work-related activities in forested areas. Industries often follow best management practices and work collaboratively with local and regional authorities to prevent wildfires by limiting operations in times of extreme fire danger, and by mitigating hazards by removing flammable vegetation and using equipment that is up to code and that meets safety standards. For example, the Government of Alberta, Partners in Protection, and the Safety Association for Canada's Upstream Oil and Gas have partnered with FireSmart to develop a wildfire prevention field guide for the oil and gas industry in Alberta [51]. Similarly, the Forest Engineering Research Institute of Canada (FERIC) also undertakes research of this nature. Baxter [52, 53] examined the effects of all-terrain vehicles (ATVs) on wildfire ignition potential at the behest of Alberta Sustainable Research Development and provided recommendations for ATV riders to bring about awareness of fire potential and mitigation strategies to prevent such ignitions.

Fire Prevention Effectiveness

Implementing and adopting wildland fire prevention programs require knowledge about what to do, an awareness of the risks involved, and often, the risks associated with not taking action. This is true for both prevention activities that focus on social change and those more technological in nature. Without the appreciation that property and lives might be in danger and at risk of loss, it is unlikely that stakeholders would be motivated to take action or to change their behavior (except in the case of arson). Similarly, attitudes and beliefs about fire risk and program effectiveness shape how individuals behave.

To what extent do wildfire prevention education programs and tools, and prevention messaging work to reduce the likelihood of fire, and to reduce damages? Furthermore, are there management practices that enhance or hinder effectiveness? This section reviews studies that examine prevention efficacy and the characteristics associated with strategy success.

Weatherill [54], in an attempt to measure success, found that out of 396 prevention programs in effect in Region 8 in the southern USA, only one had undergone formal evaluation. He also provided a taxonomy of fire prevention programs ranging from advertising to law enforcement [55]. Donoghue and Main [56] looked into factors that influenced prevention effectiveness in the early 1980s. Specifically, they considered latitude, weather, rural population density, and the effects of law enforcement. Their results showed that these factors combined accounted for half the variability in fire occurrence. Separately, however, latitude had the greatest explanatory power, weather and density only moderate, and law enforcement very little. Their recommendations were to include more detailed data to increase the predictive power of the model.

Fire prevention effectiveness can also be reliant on public agents' perceptions about their job requirements and responsibilities. Christiansen et al. [57] interviewed personnel employed by the Forest Service to assess differences between perceived and actual roles related to fire prevention and how such perceptions hindered prevention success. Barriers to success were many and included lack of time, manpower, funding, and training, and inadequate law enforcement. One of the most telling findings was the inability to define "success" and the need for better methods to determine successful prevention.

Measuring the effectiveness and benefits of wildfire prevention tends to be difficult for two reasons: an effective strategy negates a negative consequence, and it is difficult to run controlled experiments to isolate characteristics of strategies that lead to success. In cases where landscape and private property had been treated prior to a wildfire, results are varied and inconclusive. For example, a study by Graham et al. [58] revealed that fuel treatments in Colorado failed to prevent significant loss in the Fourmile Canyon Fire near Boulder, Colorado, in 2010.

⁸ Railroad Fire Prevention Field Guide: http://cdfdata.fire.ca.gov/fire_er/fpp_engineering_view?guide_id=9. Accessed May 18, 2018.

⁹ ARTC: <https://www.artc.com.au/community/fire/> Accessed May 18, 2018.

Bennetton et al. [59] conducted an economic evaluation of both bushfire prevention and suppression in Victoria, Australia. They estimated market and non-market values that would have been lost without effective wildfire prevention. Fire prevention activities considered in the analysis included fuel management and prescribed burning, and a host of infrastructure modifications from road maintenance to increased equipment availability. Return on investment was positive at about 24:1, which they estimated to be low given the difficulty in accurately measuring the value of ecosystem services.

Prestemon et al. [60] quantified effectiveness of wildfire prevention education in terms of cost savings associated with the avoidance of accidental, human-caused wildfires in the USA. Their findings indicate that wildland fire prevention education is statistically related to reductions in fire occurrence for preventable fires. Additionally, the marginal benefit from prevention education in terms of damage averted was 35 times the expected marginal cost. Ongoing research by Prestemon et al. [61] further recommended that programs focused on campfire escapes, arson, and fires lit by children would likely have the greatest positive benefits. They went on to recommend that prevention efforts on tribal lands were highly effective, but that more research was needed to determine the types of programs that would yield the best results. Regardless of land ownership, more and better data are necessary.

Abt et al. [62] focused their research on tribal lands in the USA to assess the effectiveness of fire prevention programs on accidental and incendiary fires. The researchers evaluated activities such as public service announcements, outreach activities aimed at youth, and legal enforcement. Results indicated that prevention efforts were effective at reducing the numbers of wildfires caused by campfires, juveniles, and careless fire use. Additionally, increased law enforcement helped to reduce incendiary fires. The authors also estimated the benefit to cost ratio being about 4.5:1 across all Bureau of Indian Affairs regions after controlling for biophysical effects.

Given the ideological shift from treating fuels on public lands to mitigating risks on private property, it is necessary to understand what motivates private actions. This includes (i) the role of education in increasing risk perceptions and awareness; (ii) the likelihood that stakeholders will take action; (iii) the effect of stakeholder involvement and collaboration within and across political and social boundaries; and (iv) the importance and role of effective communication. These characteristics have been examined to determine their effects on knowledge and human behavior, but are more difficult to assess quantitatively in terms of success, particularly with respect to resultant decreases in fire ignitions and/or values protected.

Heightening awareness of risk and safety has been shown in many studies to encourage human behavioral change in favor of adopting mitigation measures [37, 63–65]. Such programs work best when they are tailored to meet individuals'

needs, are site specific [66–68], provide only necessary and relevant information [69], are interactive [68], and include a clear definition of risk [70]. This latter point is highly important because risk can be interpreted in a variety of ways. Miller and Ager [71] provide useful guidance by using a definition from the Society of Risk Analysis that clearly articulates what risk means in a fire context. Such language could be adopted within prevention programming to better inform stakeholders.

Risk must also be communicated appropriately to be an effective source of motivation. Rohrmann [72] provides best practices in risk communication for disaster preparedness. Because wildfire prevention and protection require resident/public involvement, effective risk communication and education are vital and must take into consideration risk perceptions, community characteristics, perceived responsibility, and existing knowledge.

In spite of many positive findings, the relationship between risk awareness and the adoption of risk mitigating actions is tenuous. Faulkner et al. [73] looked at the role government agencies play in changing risk perceptions and encouraging people to mitigate risk on private property. They compared perceptions of risk, responsibility, and adoption in towns in Alberta, Canada, with and without fire management programs. Their findings suggested that residents gained increased risk perception and awareness of wildfire in towns where mitigation activities took place. However, government activities did not lead to greater responsibility or action taken by residents. This is a critical finding because converting knowledge to action is paramount in preventing fires. The authors speculated that it might take more time to change resident behavior, that financial barriers might exist, and that in spite of a heightened awareness of risk, homeowners believed that firefighters would be able to protect them.

Alternatively, Jarrett et al. [74] looked at private nonindustrial forestland owners in the southern USA to identify their level of awareness and likelihood of adopting prevention strategies. While they recognized that awareness does not lead to adoption, they found significant differences in socioeconomic characteristics that indicate whether individuals are more or less likely to adapt. Specifically, awareness, risk perceptions, and preferences for intervention and information were dependent upon gender, experience with wildfire, knowledge, and desire for assistance. Other studies also reveal that educational programming is dependent upon socio-demographics and diversity [75, 76]. These findings are promising, but are site specific and cannot be generalized given the complexity of human nature and localized factors.

Champ et al. [77] examined the relationship between awareness and adoption using a survey of homeowners in Colorado. The authors develop two models: one that examines wildfire risk-mitigation behavior as a function of risk perception and the second involved wildfire risk-mitigating behavior and perceived risk simultaneously. Their results show that

perceptions and risk-mitigating behaviors are jointly determined. In other words, as individuals learn more about wildfire risk, they also learn more about how to mitigate that risk.

How information is disseminated can also influence the effectiveness of knowledge transfer. Research focusing on communication regarding wildland fire is extensive across all aspects of management. Additionally, there is a large body of research examining effective communication forms and the role of media. Early on, the importance of communication was recognized as key to encouraging individuals to heed fire prevention messaging. Doolittle and Welch [78] studied intensive personal contact as a way to disseminate wildfire prevention messaging. Their results showed a 55% reduction in human-caused wildfires over a 5-year period in Beauregard Parish, Louisiana.

Communication can take many forms, but requires key elements: planning; unidirectional and interactive approaches; a focus on issues of local interest; and meaningful interaction that builds trust [75, 79]. Absher and Vaske [80] further explored the role of “trust” as it influenced homeowners’ mitigation behaviors. They looked specifically at trust in agency competence and trust in agency information. Their results showed only weak influence on mitigating behaviors leading them to conclude that trust is likely mediated by other factors.

Earlier work by Doolittle and Lightsey [81] made specific mention of how messaging is delivered to the public by wildland fire agency personnel. Individuals responsible for working with the public to enhance wildfire prevention were more successful as measured by reduced wildfire frequency when they were people-oriented rather than forest-oriented. Research suggested that adopting a systematic, professional approach to communication helped to convey fire prevention messaging more effectively.

Media can be an effective tool to disseminate messages according to Johnson et al. [82]. The authors examined the role of the media in communicating best practices around defensible space. They suggest that during large spikes in wildfire activity, the media could be used to communicate the importance about defensible space, and the fact that it can be accomplished with a full range of actions, rather than just vegetation management.

Studies have also examined what happens when communication fails. Eriksen and Prior [83] acknowledge the gap between an individual receiving information and then acting on it in a study of Australian bushfires. They argue that there is miscommunication of wildfire risk because fire personnel and volunteers are not mentally prepared to convey information in situations where wildfire threats are realized. Similarly, Stidham et al. [84] compared management styles used in two communities that experienced catastrophic wildfires: one community experienced clear communications while the other did not. There was a marked difference in preparedness based on trust and the relationships that began prior to

ignition. Results speak to the importance of collaboration and communication for effective management during a crisis.

Communicating through networks is another area of increasing interest. Social capital, defined to be networks of relationships, can be an important component of community preparedness. Jakes et al. [85] studied several U.S. communities in Florida, Minnesota, and Oregon and identified five social elements that enhance a community’s preparedness: (i) the ability to function collectively as a community; (ii) having the necessary knowledge and skills to engage in preparedness activities; (iii) having knowledge and skills associated with heritage, experience, sense of place; (iv) understanding landscape factors related to sense of place, logistics, and ownership; and (v) agency involvement for collaboration, expertise, and skills. They did not, however, consider economic factors such as financial capacity or funding.

Social capital is positively related to both economic and non-economic benefits. Agrawal and Monroe [86] used a survey approach to assess social capital specifically described as respondents’ perceptions of the quality of relationships with their neighbors, members of the community, and the degree of community participation. They found that people who perceive greater social capital in their communities were more likely to engage in activities that build knowledge and skills to prevent wildfire, create defensible space around their property, and to communicate and collaborate with others. Also, regarding the power of neighbors and communication, Johnson et al. [80] conducted interviews in six diverse communities across the USA and concluded that neighborhood organizations can be effectively used to mitigate fire risk. They note that there must be collaboration among stakeholders and that solutions must fit the needs of the community.

The power of community collaboration has been shown to positively enhance homeowner wildfire preparedness. Akama and Ivanka [87] look more deeply at how a community is defined, who participates, and potential challenges to participatory actions. They recognize the importance of sub-groups of common interests within communities, and the power of informal networks to increase engagement. In later work, they also note that awareness does not always lead to action [88], demonstrating that the typical one-way communication of wildfire risk from authorities to the public is not enough to empower homeowners to take control of their own preparedness. The authors suggest new methods reliant on social interactions to more effectively enhance preparedness activities through knowledge sharing and the establishment of fire learning networks.

Many authors have studied the importance of social interactions in the WUI. Brenkert-Smith et al. [89] look specifically at the relationship between information sources and social interactions. They classify fire information as expert and non-expert, and social interactions as formal and informal, and generic versus fire-specific. Results of their research indicate

that social amplification (i.e., sharing of information) plays a role in determining individuals' perceptions of wildfire risk. Additionally, they find that perceived risk is affected by vegetation on neighboring properties. Their results can be useful to wildfire managers to improve the understanding of how risk is communicated among different stakeholders, and that the public responds to risk in a way that depends on the social context.

Finally, Dickinson et al. [90•] explore the effects of social interactions on beliefs pertaining to wildfire risk (probability and consequences of damages) and risk mitigation options (benefits/costs and efficacy) that lead to adoption of risk mitigating behavior. They focus on behaviors that mitigate risk through fuels reductions on and around private property and structural modifications to houses. Results show a positive relationship between social interactions that are fire specific, and risk perceptions around the probability of fire. This relationship then positively influences vegetation management. However, this is not true for structural modification. The authors summarize potential pathways that shape beliefs and mitigation behavior (e.g., information transmission, social amplification, risk interdependency, and social norms and capital), but note that correlation is not causal, and that the results are not generalizable.

The effectiveness of fire prevention efforts not only depends on how information is communicated, but when and at what age it is communicated. Mockrin et al. [91] looked at individual decisions to rebuild or relocate after a wildfire. They recommend using the window of opportunity that occurs after a catastrophic fire to promote adaptation in the community affected, and other WUI communities. It is also important to start fire prevention education while people are young. Ballard et al. [92] examined 50 Wildfire Education programs for Youth (WEY) and found that not only do they focus on wildfire prevention and safety, but they now reflect changing attitudes toward fire that are based on ecology and management. Children are taught in many programs about fire safety and preventing unwanted fires, but also learn that fire is an important ecological process.

Butry et al. [93] conducted a comprehensive analysis of the effectiveness of a range of wildfire prevention efforts targeted to specific audiences at different junctures. They evaluated general public service announcements broadcast through various media, home visits, presentations, brochure distribution, and community-based wildland hazard assessments in Florida. They also considered timing of prevention activities noting that most occurred in March, April, and May, which coincides with peak wildfire activity. Their results indicated that benefits of fire prevention and education activities far outweigh the costs, and depending on weather, fire history, management activities, and season, altering the timing of prevention activities will produce public benefits in terms of cost savings.

Discussion

In its simplest form, wildfire prevention means to keep something from happening (i.e., fire occurrences). In the case of wildfire, it is to prevent unwanted fires from occurring, and if they do occur, to minimize spread, and reduce and mitigate damage. Research in this area is broad and informs all aspects of prevention from estimating the likelihood of ignition by location, time, and cause to evaluating socio-demographic factors that influence effectiveness. Prevention strategies are built on many approaches informed by research from a range of disciplines that span the natural sciences (e.g., biology, ecology), social sciences (e.g., sociology, criminology, economics), and health sciences (e.g., psychology and medicine). Additionally, fire prevention most often relies on legal and regulatory approaches that seek to influence human behavior through positive or negative incentives.

McCaffrey et al. [94] provide a comprehensive assessment of the social science research (excluding economics) for wildfire management in general. They conducted a literature review and subsequent workshop to identify 12 overlapping themes. Their results indicate that there is public support for mitigation efforts prior to unwanted wildfire occurrences and that there exist effective methods to address social complexities regarding risk awareness, yet gaps remain. They conclude that because fire management is largely reliant on changing human behavior, understanding social implications of wildfire occurrence will continue to be important to wildland fire management in general. This supports the general findings within the prevention literature that because most fires are human-caused, a reduction in the number of unwanted fires cannot change without behavioral change.

Overall, research indicates that support for community initiatives and programs aimed at risk reduction in the WUI are dependent on many diverse factors—fire experience and risk awareness, motivation to protect family/friends, and encouragement from government [95, 96], commitment to collaboration and shared values [97], and an understanding of scale and the ability to be proactive [98]. Research has identified best management practices that could increase support for prevention measures and start to lead to greater adoption. While awareness is necessary, it is not always sufficient to stimulate action. Furthermore, programs and strategies that are flexible and allow communities to tailor solutions to local conditions and community needs seem to be more successful in fostering engagement and positive results.

Communication and education that is interactive provide greater opportunities to effect change in attitudes and beliefs about fire prevention. Education designed to target demographics according to socio-economic differences is more likely to be effective given differences in population cultures, attitudes, and beliefs. This is supported by spatio-temporal research on ignition likelihood, for example. Efforts to

increase capital in all forms (social, cultural, financial) can support decision-making and enhance implementation strategies. Social capital can be improved through greater collaboration among stakeholders as well as knowledge transfer through education and experience. The very process of building social capital helps to strengthen communication among all stakeholders and might increase resilience in the event of a fire thus better enabling communities and individuals to deal with crisis. Assistance and support (financial and physical) can encourage action and motivate individuals to take the necessary steps to modify structures and landscapes on private land. Furthermore, strategies that are collaborative include several stakeholders, and cross jurisdictional boundaries will lead to more comprehensive solutions. Fire prevention efforts are more likely to be successful if stakeholders take a multi-faceted approach.

Overall, fire prevention has moved from an attempt to remove all fire from the landscape, to one where communities and agencies sometimes use and embrace fire as a natural ecological process. Fire prevention strategies have shifted from treating primarily public lands to a greater focus on treating private lands through vegetative manipulation and structural modification. With this comes a partial shift in responsibility from the public sector to the private sector, and converting knowledge into action.

There has been a wealth of research conducted on risk perceptions, attitudes, and beliefs regarding wildland fire and how or whether wildfire prevention education can influence changes in human behavior. While some research shows that stakeholders who believe there is a risk of wildfire are more likely to take action to adopt protective strategies, others are indeterminate. Evaluating the degree to which such strategies work and their cost effectiveness is also difficult to generalize, and indeed, there is no single agreed upon quantitative measure of success. Many economic studies reviewed here indicate that the benefits of education outweigh the costs by a significant margin; however, much of the social science research does not evaluate cost effectiveness. Success could be considered enhanced knowledge and changes in attitude, or adoption of specific behaviors or programs, none of which guarantee a reduction in fire ignitions or damage and loss.

Ultimately, research still seeks answers to the same questions that were asked in the 1960s. Folkman [2] talked about the need for enhanced fire prevention programming given an increasing population and concomitant demands on forest resources. He noted that education and fuels treatments were important but that behavioral studies were required to shed greater light on how to modify human behavior and better achieve prevention objectives. His research focused on linking attitudes and knowledge regarding fire prevention and wildland fire risk, with socio-demographic backgrounds and fire prevention behavior—objectives of many later studies reviewed here. He stated

that “attitudes are elusive, frustrating things to research. However, their considered importance in determining human behaviour is such as to force the attempt” [2]. His conclusion regarding the delivery of better prevention programming was that “it remains to be seen to what extent such changes will result in changes in the actual behaviour of wildland users and consequent reductions in fire starts” [2]. Fifty years later, definitive connections have not yet been made between targeted prevention efforts and changes in human behavior, and more importantly, whether changes in behavior lead to success. Furthermore, success itself seems to be an elusive outcome: one that is difficult to measure and reliant on continued efforts to educate an ever-changing population.

Research over the last several decades has made considerable advances in our understanding of aspects related to the likelihood of fire occurrence for both natural and HCFs. When it comes to the effectiveness of programming to prevent fires, most research uses a single disciplinary approach. Because the majority of wildfires worldwide ignite due to human influence, the problem is tremendously complex and will continue to benefit from investigation that uses interdisciplinary and holistic perspectives. Similarly, while research has identified many key components that are likely to improve prevention efforts, results are largely contextual and depend on local factors making it difficult to generalize. More research will be required to assess the complexities of human behavior in an effort to prevent unwanted wildland fires and subsequent loss.

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Conflict of Interest The author declares no conflicts of interest.

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